

COMPARISON OF THE FUNDAMENTAL PARAMETERS OF AP STARS HD 120198 AND HD 220825

DERIVED FROM INTERFEROMETRY AND SPECTROSCOPY

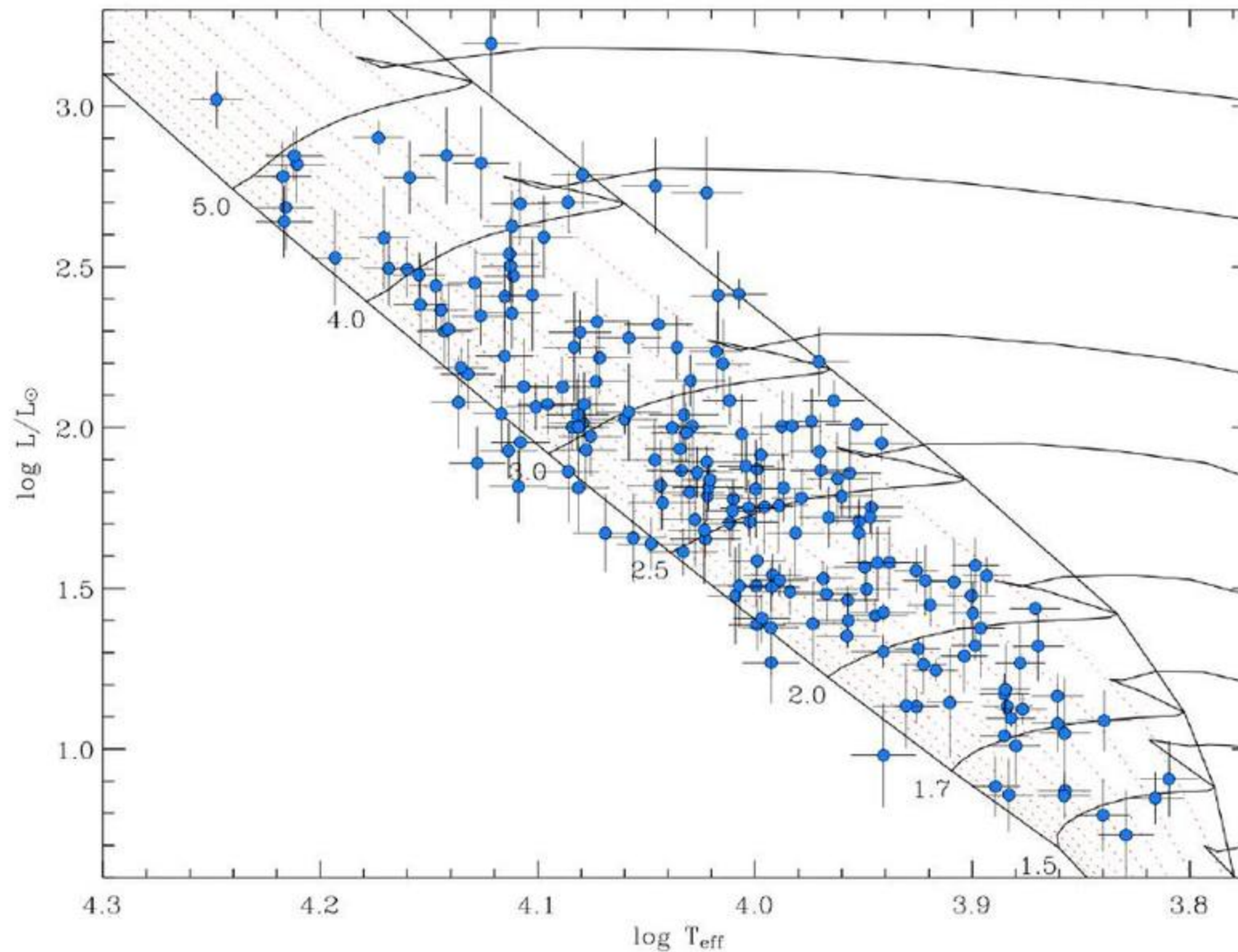


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MAGNETIC CP-STARS



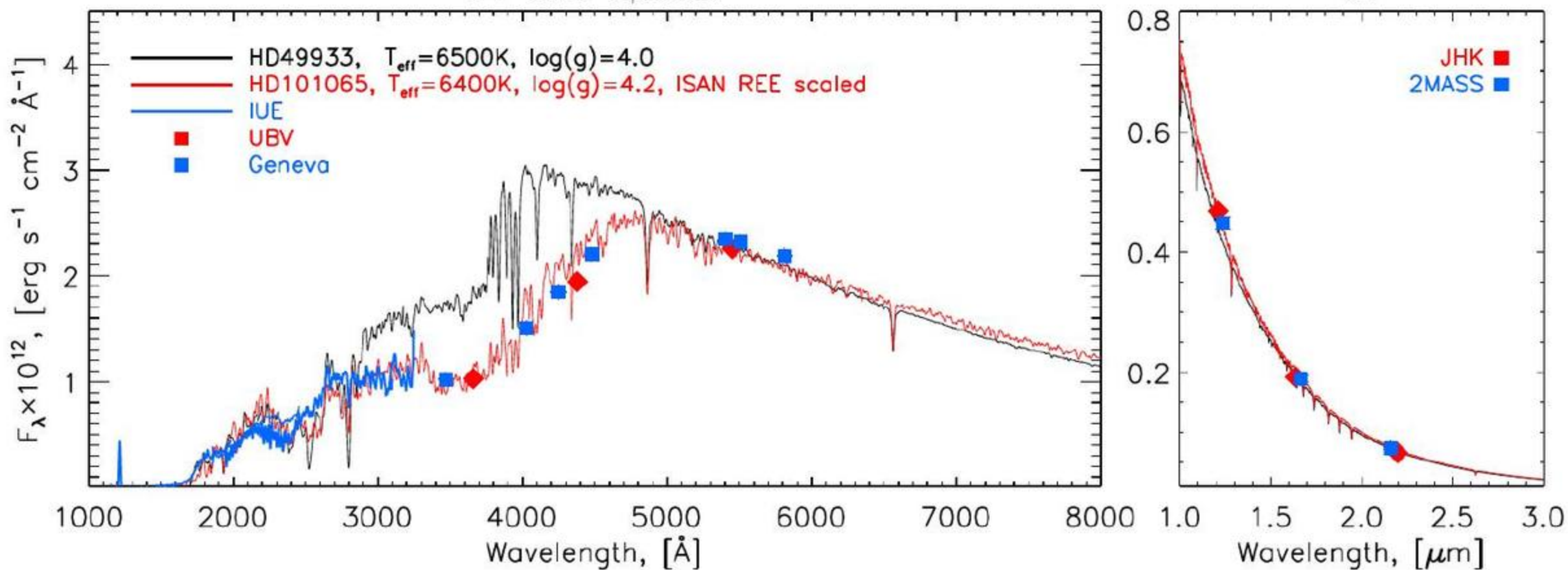
Ap-stars on the Hertzsprung-Russell diagram.

MAGNETIC CP-STARS

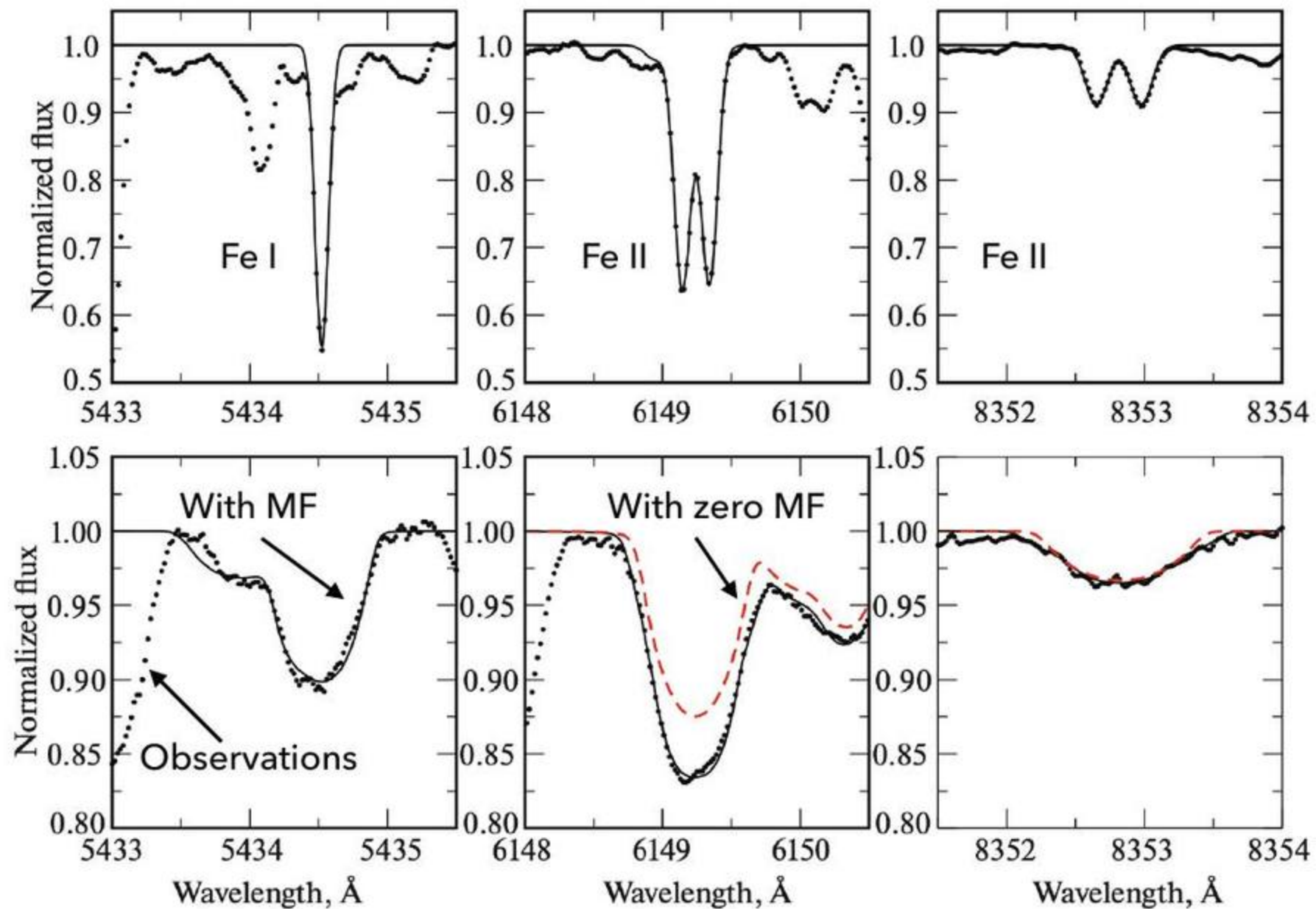
Przybylski Star HD 101065

UV and optical

IR



MAGNETIC CP-STARS

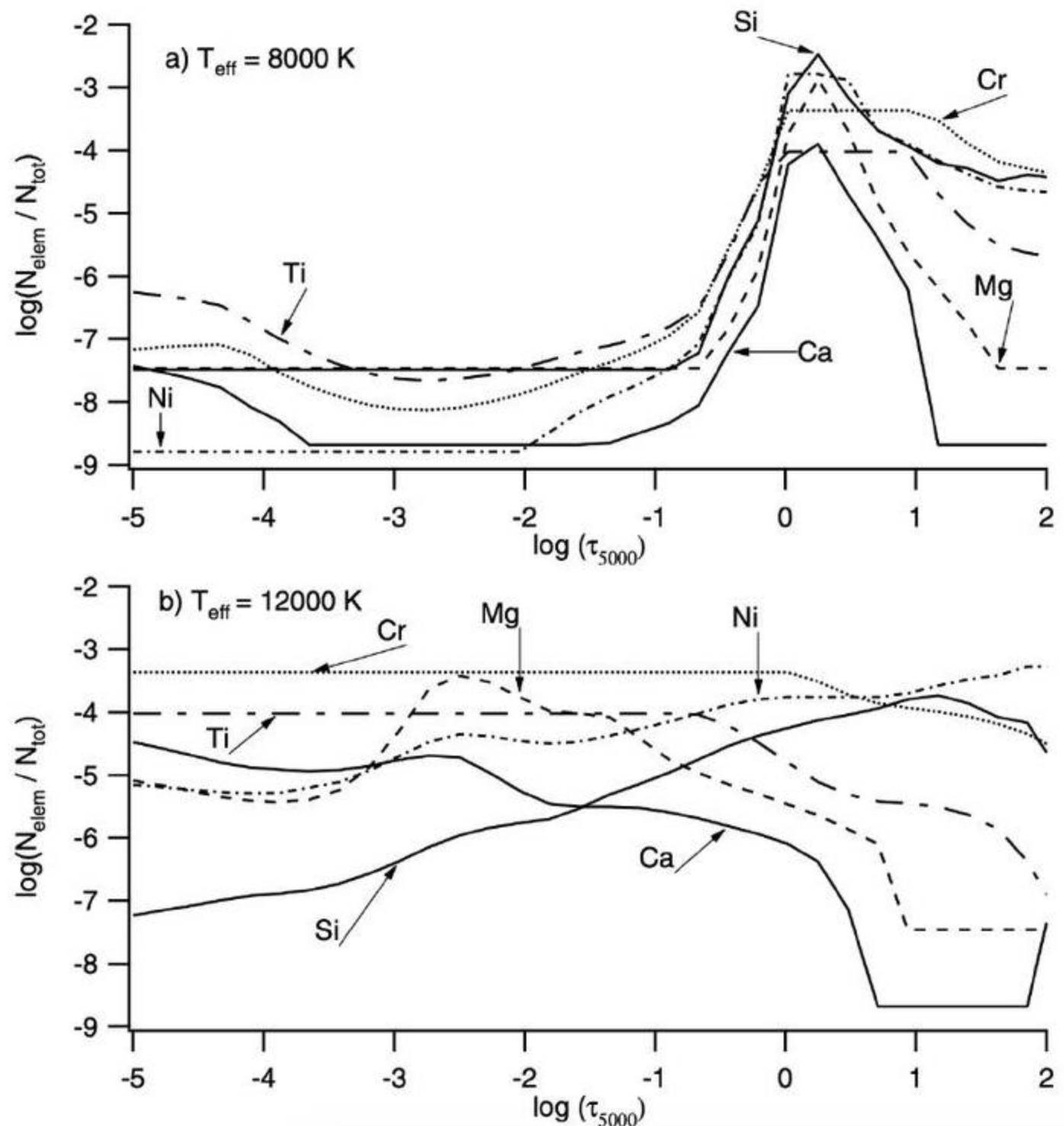


STRATIFICATION OF ELEMENTS

The abundance of the elements Mg, Si, Ca, Ti, Cr and Ni as a function of optical depth for models with $\log g = 4.0$



Abundance jump decrease with increase of temperature and move to upper layers of atmospheres



To check an accuracy of massive Ap stars parameter estimates by indirect methods (photometry and spectroscopy) a special program of interferometric observations of Ap stars was initiated. It followed up by the corresponding spectroscopic analyses

AIMS OF STUDY

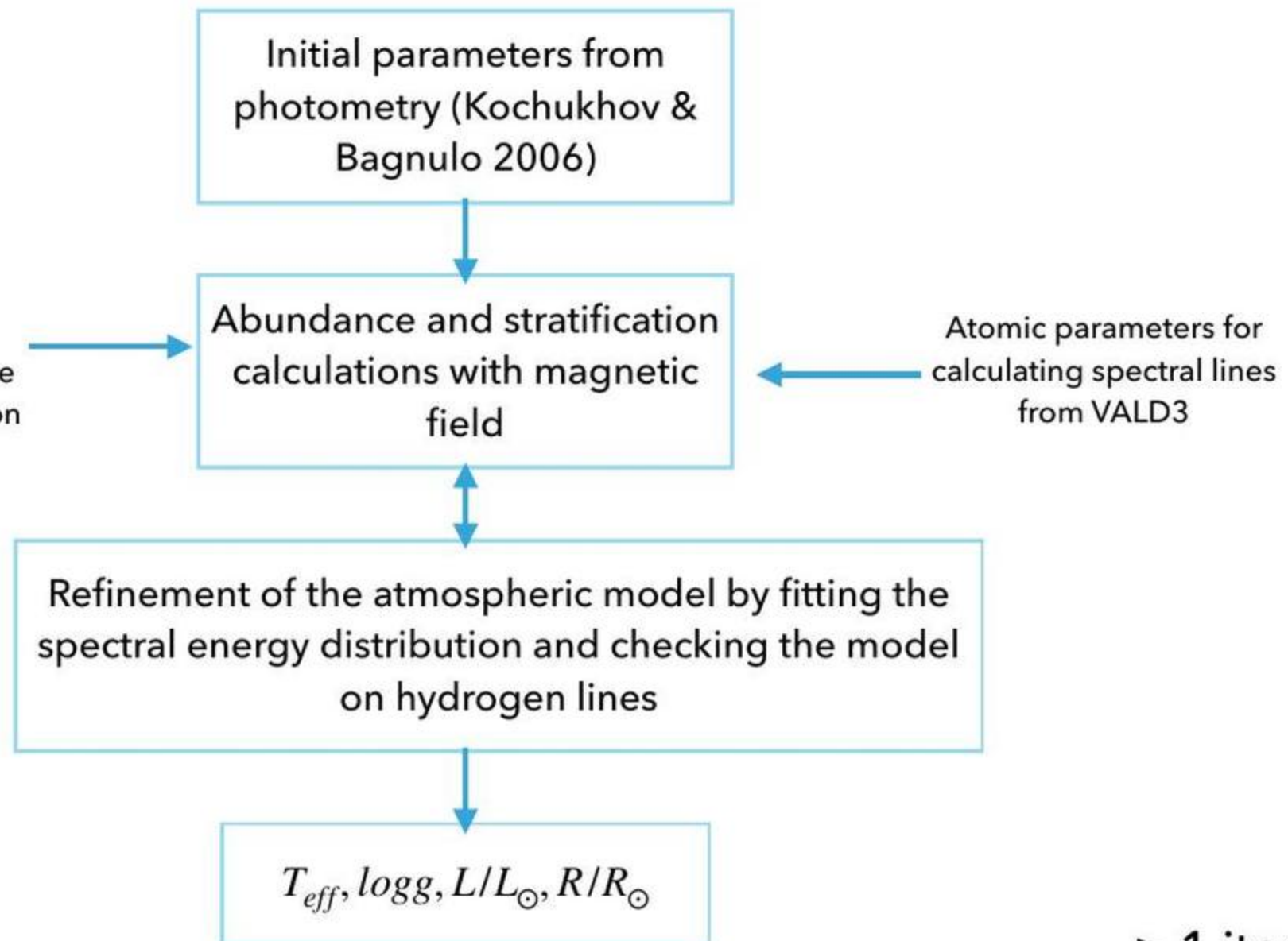
- ▶ Completion of the program of accompanying fundamental parameters determinations by means of spectroscopy and interferometry. HD 120198 and HD 220825 are ones of the last stars from the sample.
- ▶ Accurate determination of fundamental parameters by spectroscopy.
- ▶ Comparison between the parameters (radius, luminosity and temperature) derived by different methods: direct – interferometry; indirect – spectroscopy and photometry.

DETERMINATION OF FUNDAMENTAL PARAMETERS BY SPECTROSCOPY

Iterative method

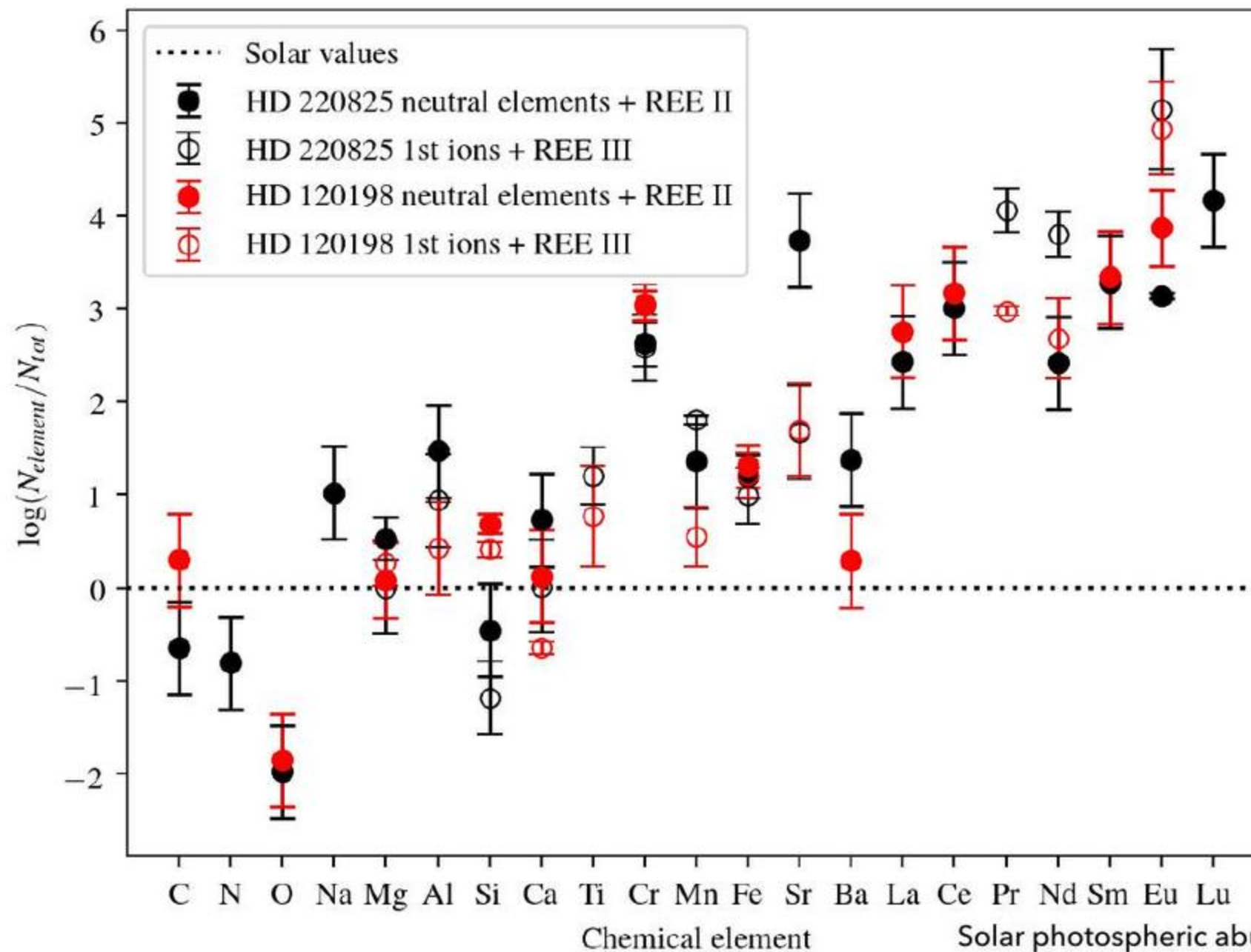
OBSERVATIONS

- ▶ Spectra of HD 120198, HD 220825 - ESPaDOnS ($R = 68000$, $3700\text{--}10000\text{ \AA}$)
- ▶ Photometry/ Spectrophotometry - UV region (TD1, IUE), optic range (Adelman, Johnson), IR region (2MASS)

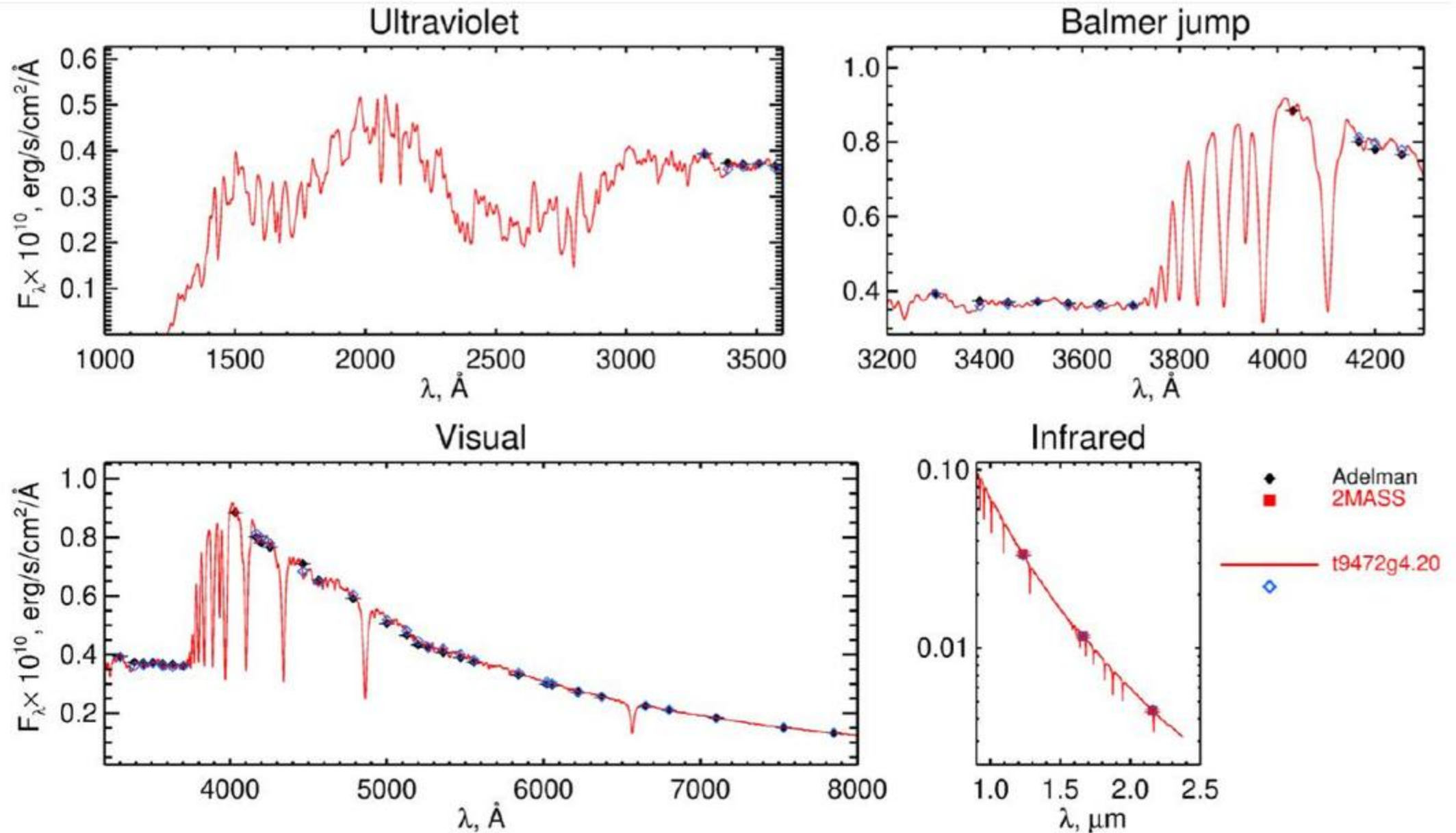


> 1 iterations

ELEMENTAL ABUNDANCES IN THE ATMOSPHERES OF HD 120198 AND HD 220825

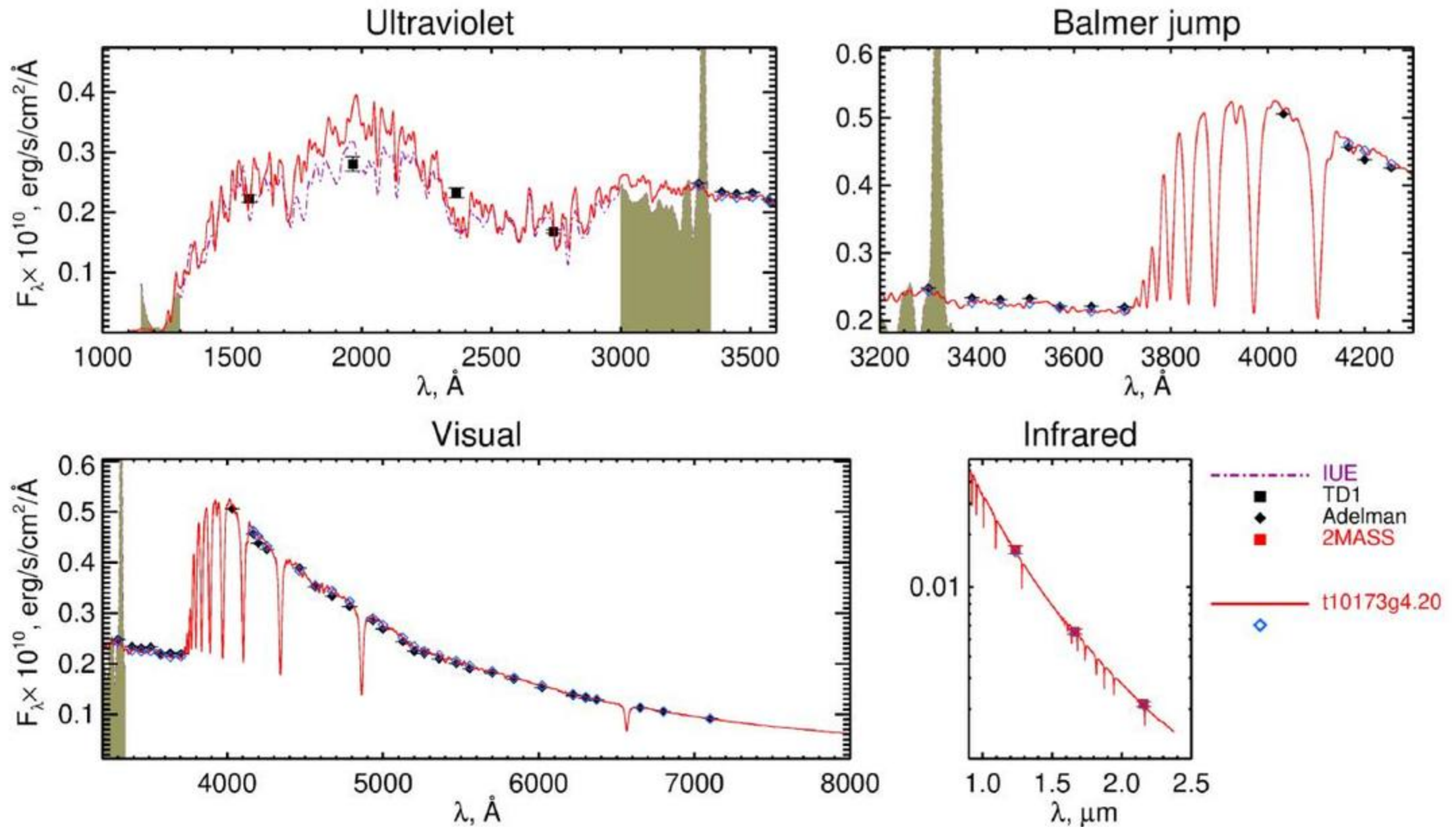


Solar photospheric abundances were taken from Asplund et al. (2009); Scott et al. (2015b, a); Grevesse et al. (2015)

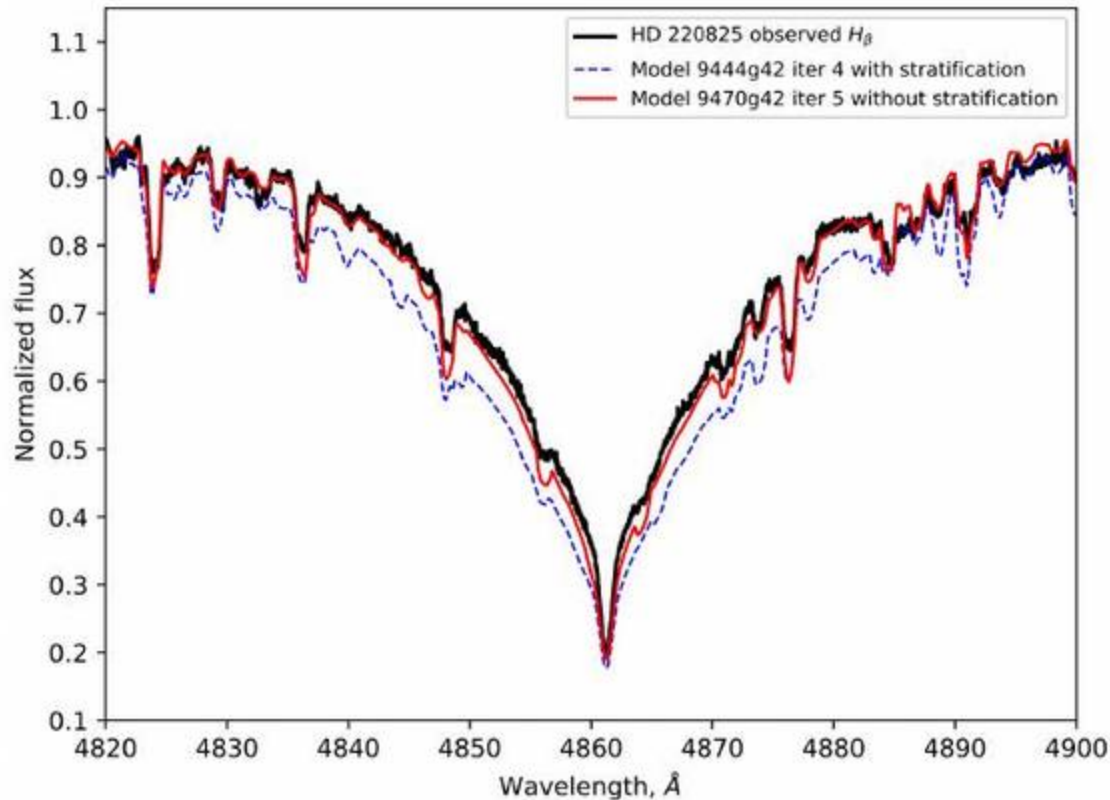
DETERMINATION OF T_{eff} AND $\log g$ HD 220825

He-weak model

5 iterations

DETERMINATION OF T_{eff} AND $\log g$ HD 120198

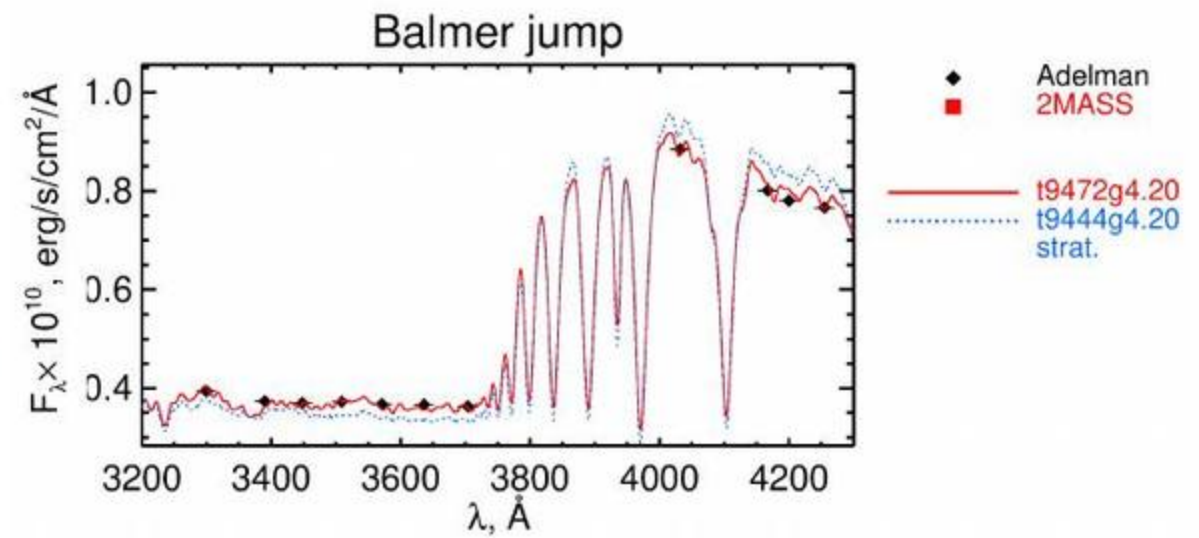
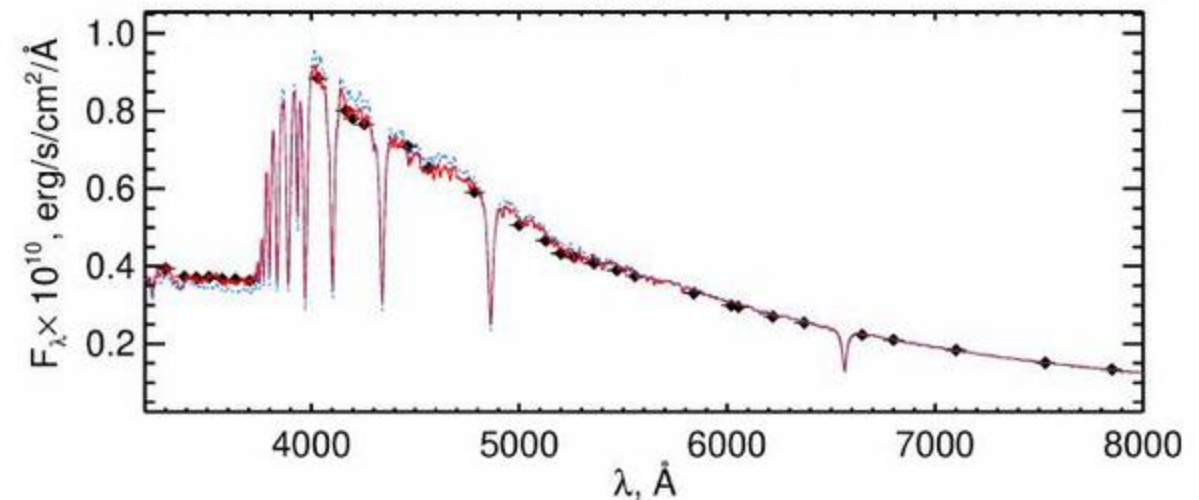
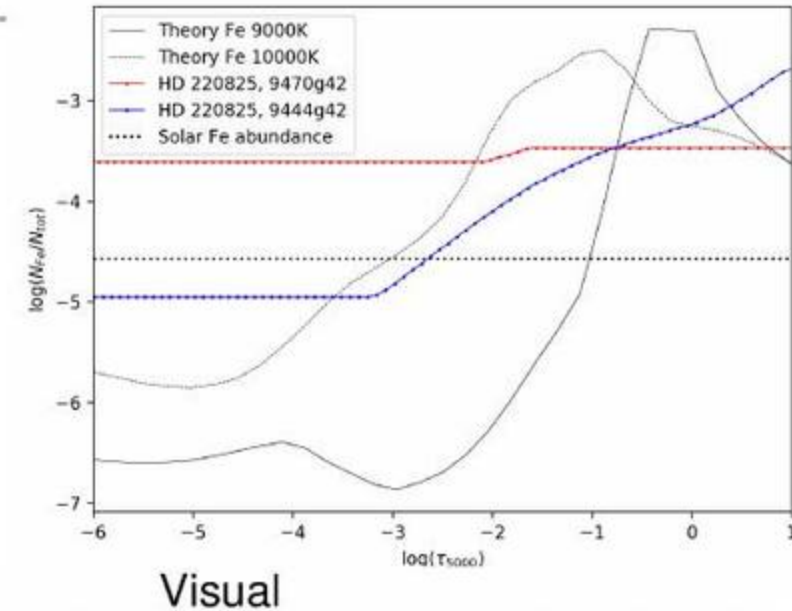
STRATIFICATION IN ATMOSPHERES



High T_{eff} and $v_e \sin i$

HD 120198 10162 K,
56.0 km/s, 1.6 kGs

HD 220825 9470 K,
38.4 km/s, 2.0 kGs



Difficult to correctly describe stratification:

High $v_e \sin i$ → lines are strongly blended

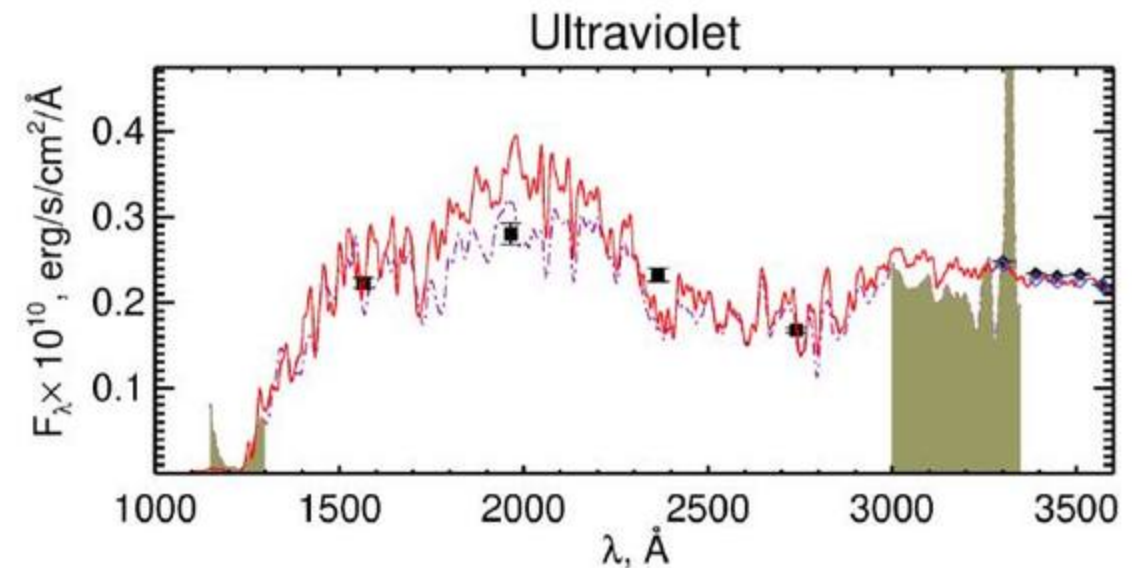
High T_{eff} → abundance jump decrease
and move to upper layers of atmosphere



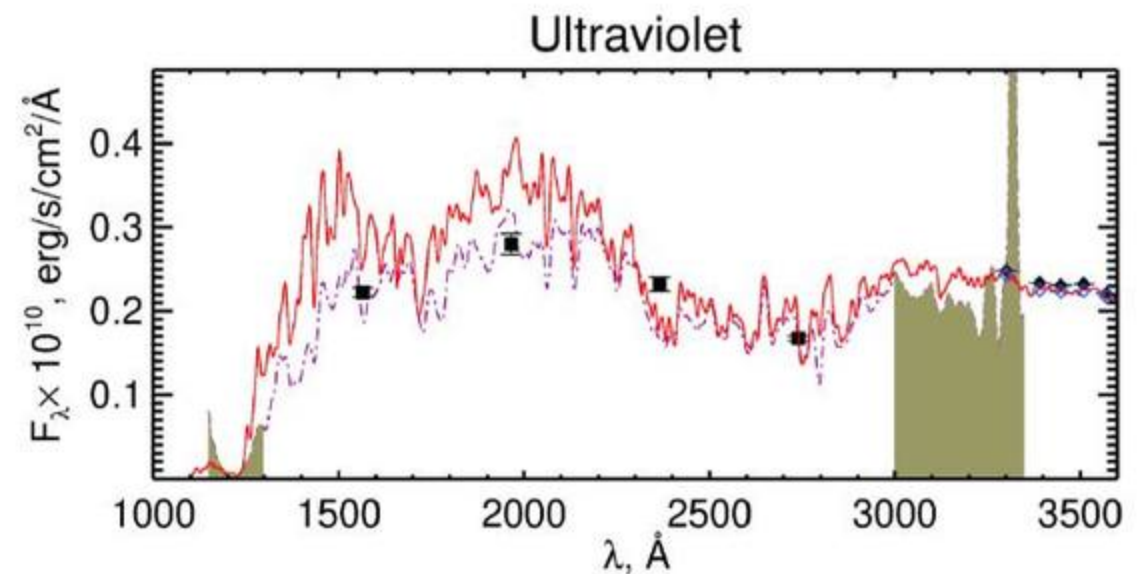
Models atmosphere without stratification

SI INFLUENCE TO SED IN THE ATMOSPHERE OF HD 120198

Model t10174g4.20
Si abundance -3.82



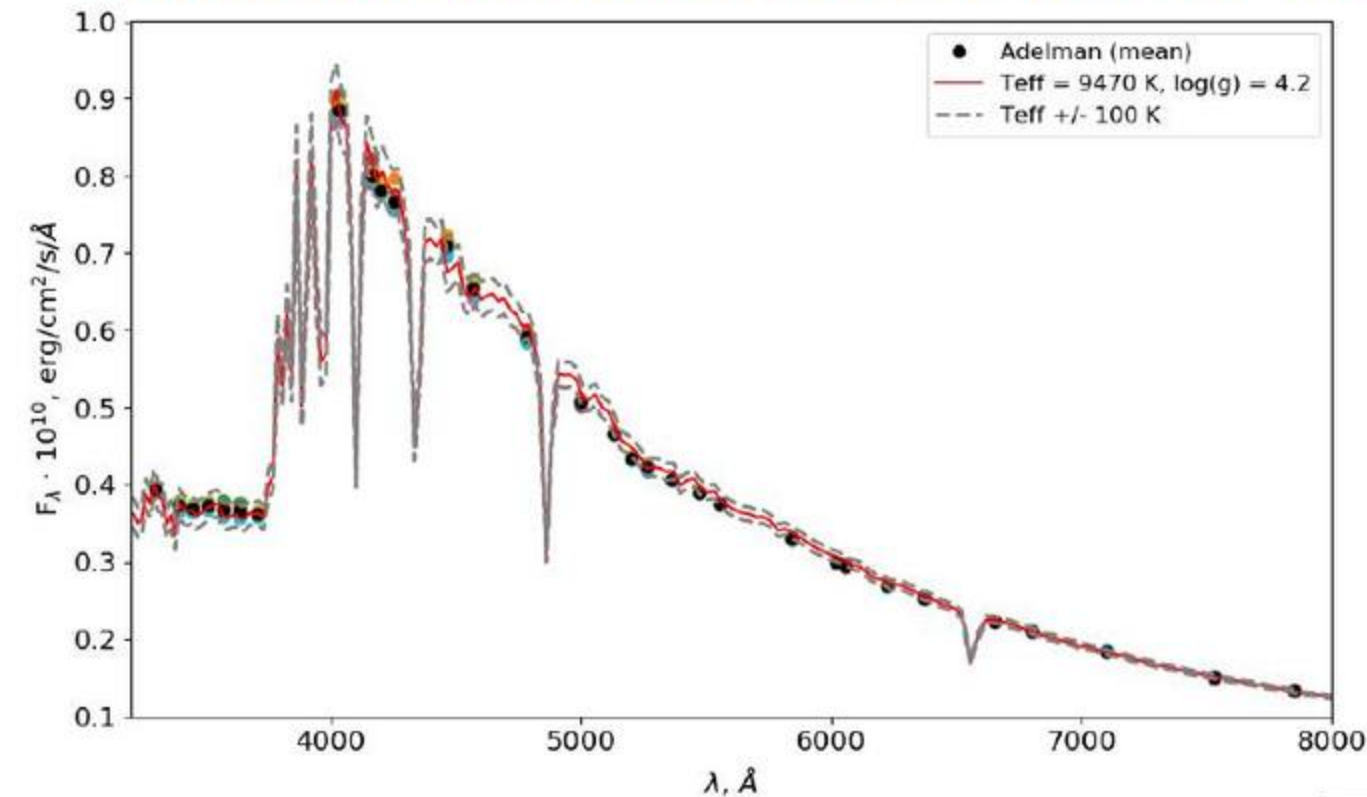
Model t10225g4.20
Si abundance -4.86



Decreasing the Si abundance by 1.0 dex leads to an overestimated flux in the region of 1500A

----- IUE
■ TD1
◆ Adelman
■ 2MASS

SURFACE ABUNDANCE INHOMOGENEITY OF HD 220825 AND ITS INFLUENCE ON SED

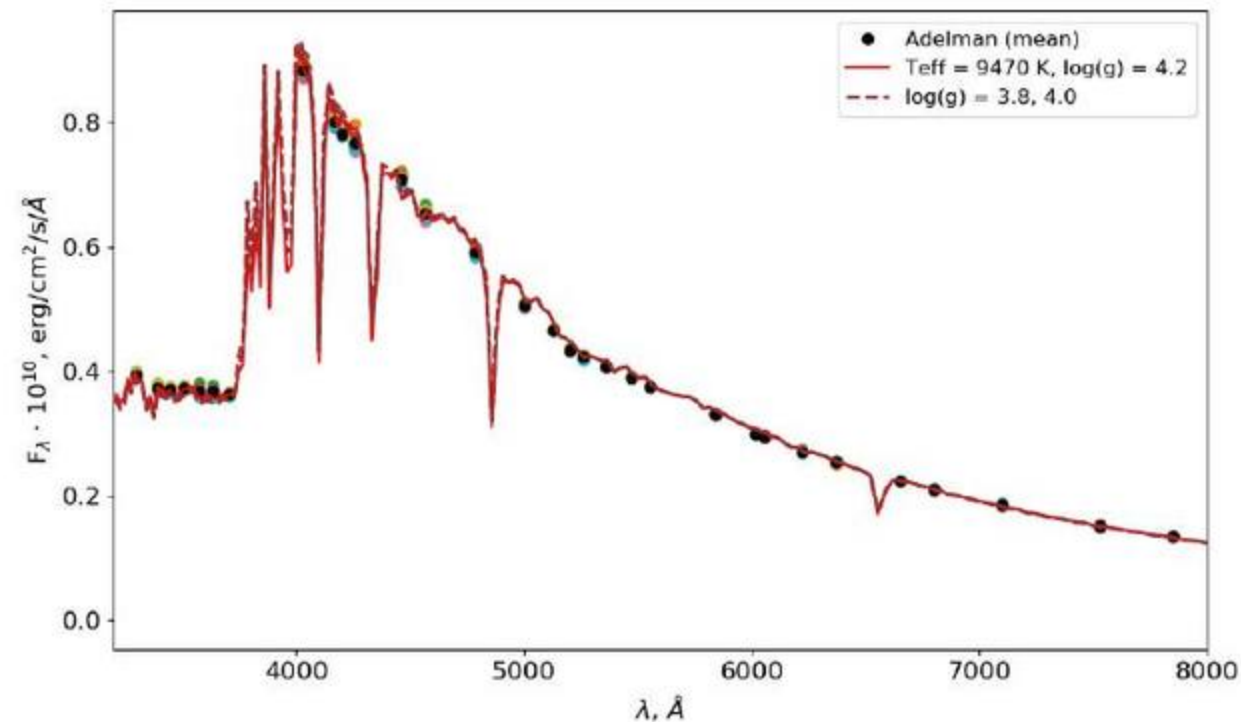


$\text{Prot} = 1^{\text{d}}.418$

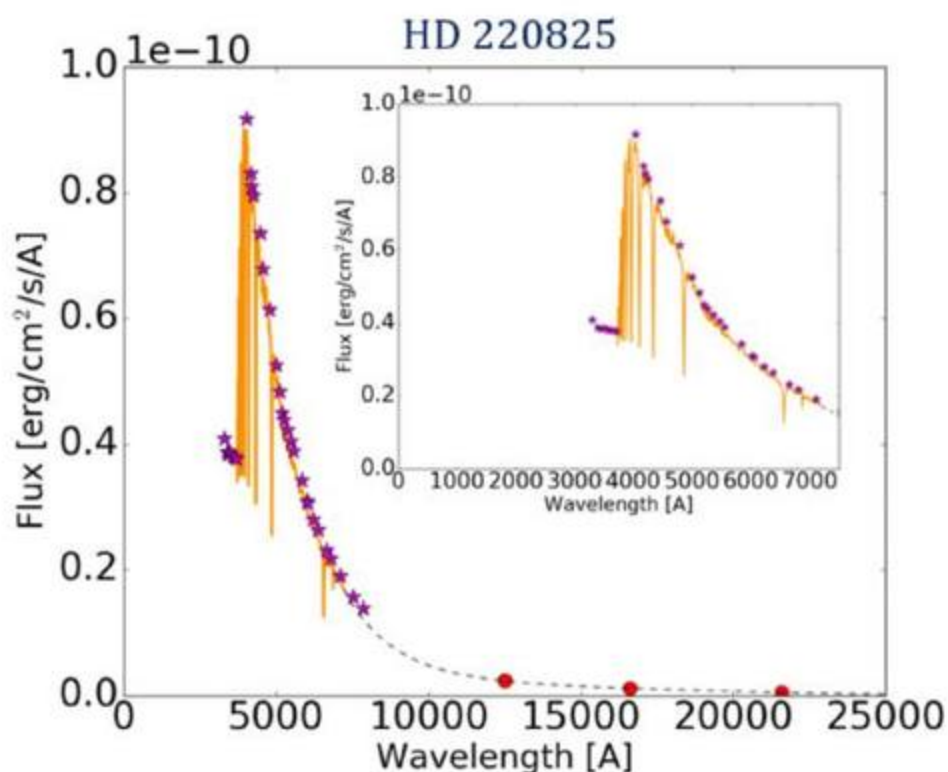
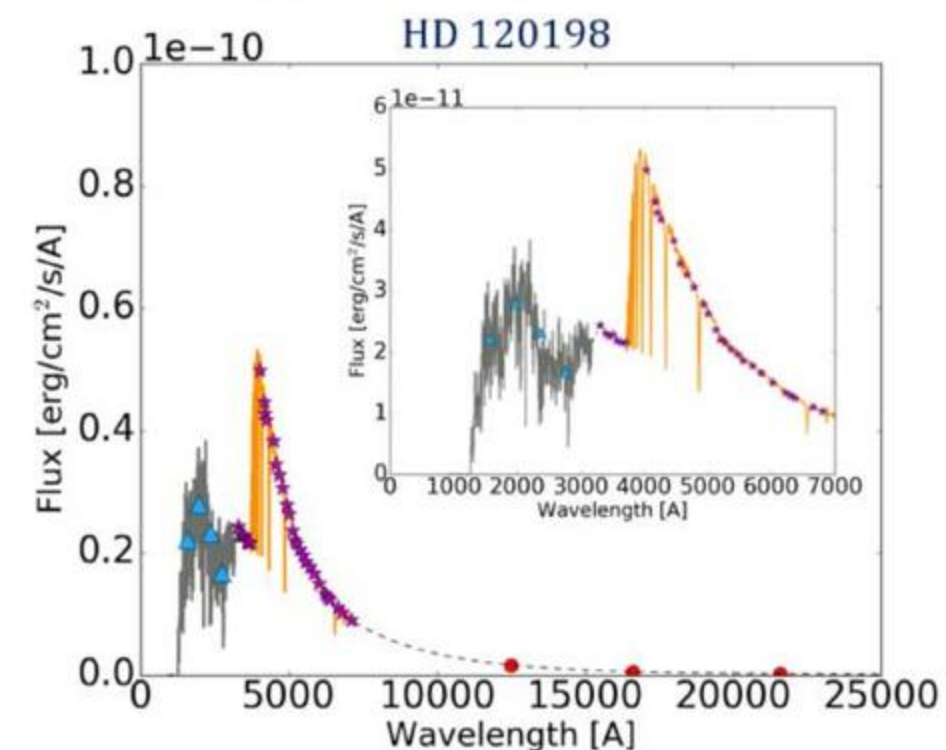
HD 220825 have the spotted surface. Particularly a chromium gradient is observed on the surface by 2 orders of magnitude and this may affect the SED (Ryabchikova et. al. 1996)

Flux variability corresponds to $\pm 100 \text{ K}$ of T_{eff} which is typical error in spectroscopy

The change of $\log g$ has little effect to SED



DETERMINATION OF FUNDAMENTAL PARAMETERS BY INTERFEROMETRY



Benchmark sample of 14 Ap stars
were observed by interferometry

f_{bol} and θ_{LD} are from observations



$$R/R_{\odot} = \frac{\theta_{LD}}{9.305 \times \pi_p}$$

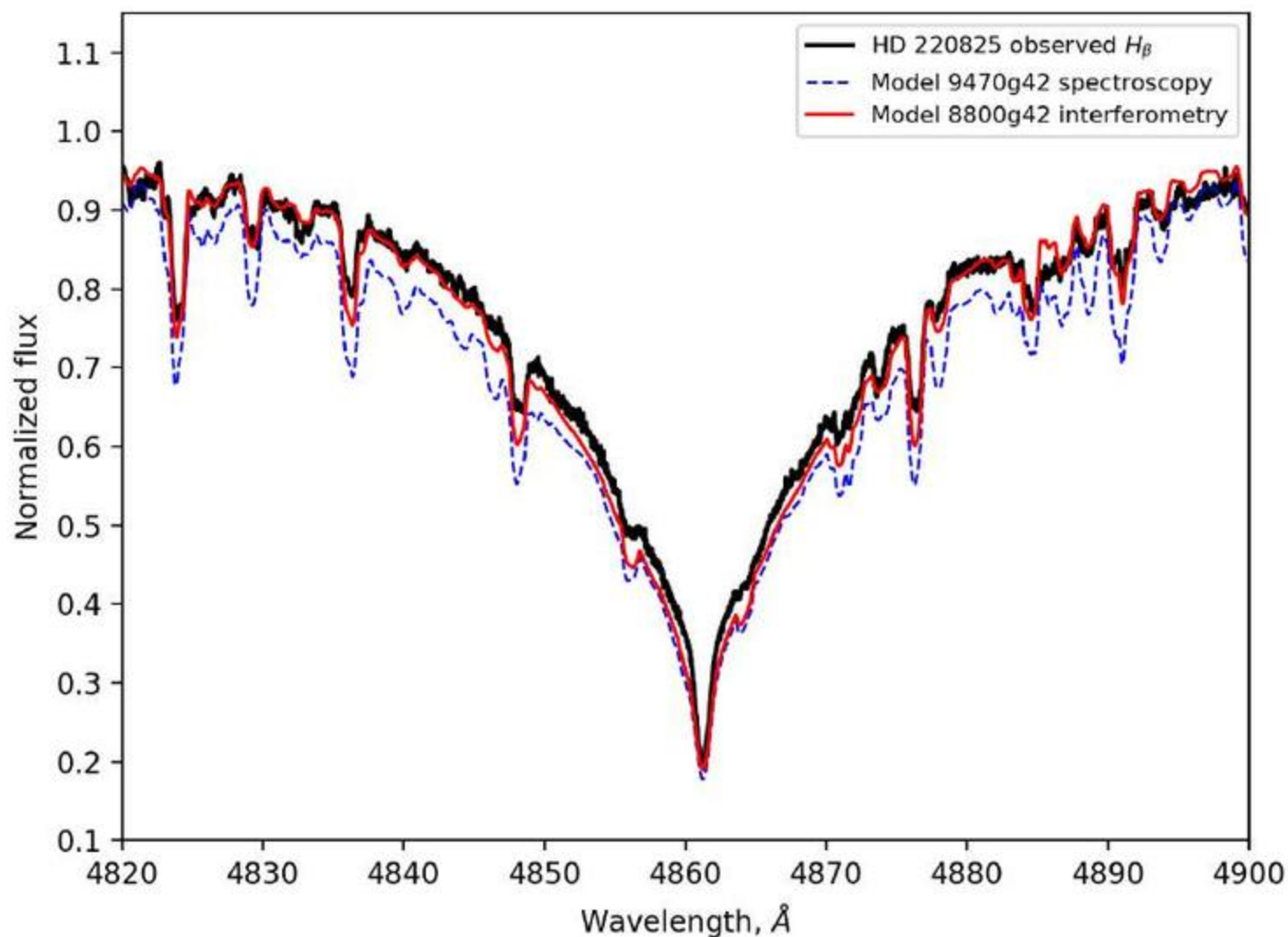
$$T_{eff} = \left(\frac{4f_{bol}}{\sigma \theta_{LD}^2} \right)^{1/4}$$

$$L/L_{\odot} = 4\pi f_{bol} \frac{C^2}{\pi_p^2},$$

where C - parsec to cm conversion
(3.086×10^{18}), π_p - parallax of the star

COMPARISON WITH INTERFEROMETRY

Fundamental parameters of 18 Ap stars were obtained by spectroscopy

 H_β IN HD 220825

HD 120198 HD 220825

B_s , kGs	1.6	2.0
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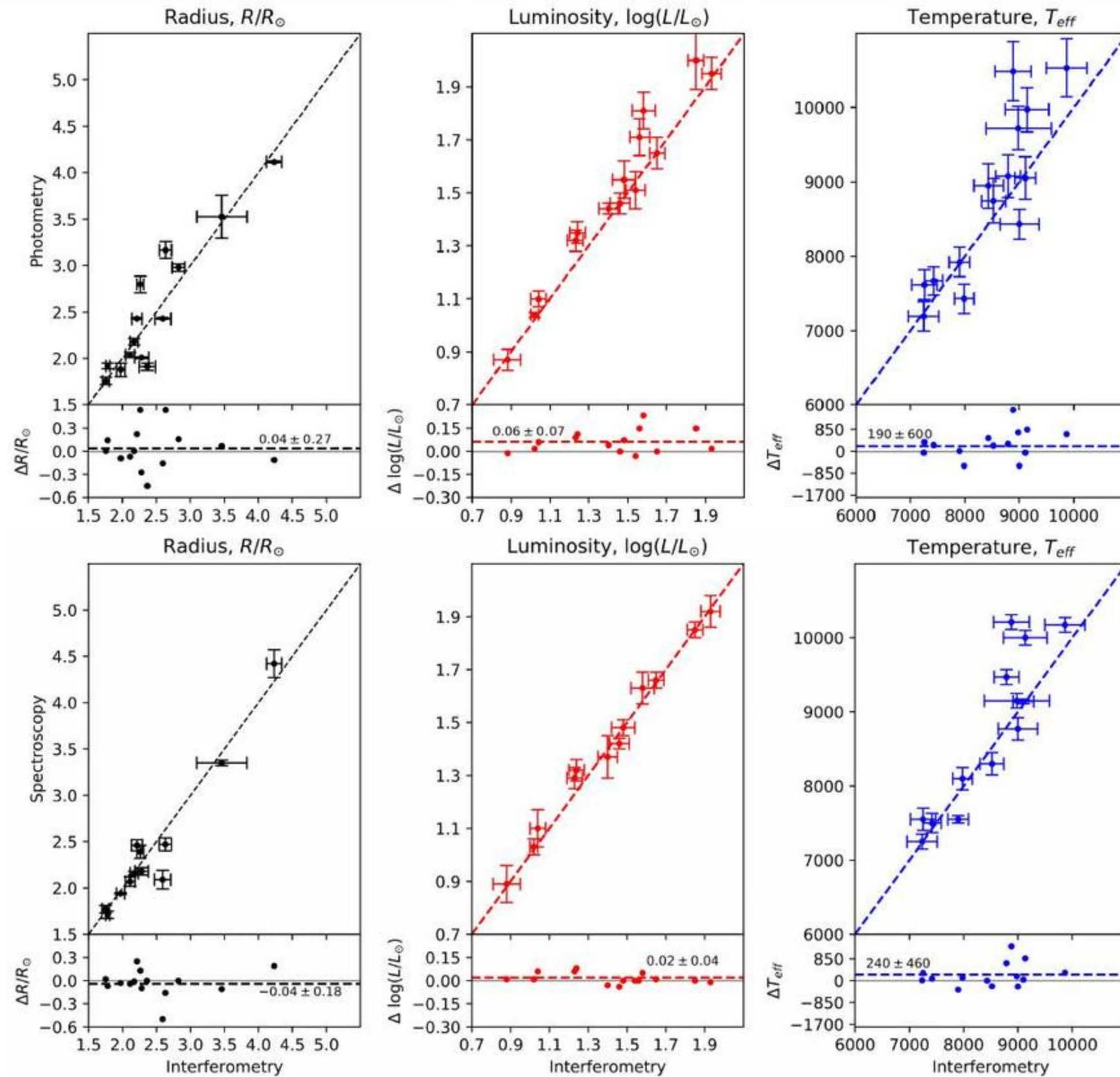
SPECTROSCOPY

T_{eff} , K	10162 ± 100	9470 ± 100
$\log g$	4.20	4.20
$R/R_\odot$	2.18 ± 0.04	1.71 ± 0.04
$\log(L/L_\odot)$	1.66 ± 0.03	1.32 ± 0.04

INTERFEROMETRY

T_{eff} , K	9865 ± 370	8790 ± 230
$R/R_\odot$	2.28 ± 0.10	1.78 ± 0.03
$\log(L/L_\odot)$	1.65 ± 0.04	1.24 ± 0.04

K. Perraut, M. Cunha, A. Romanovskaya, T. Ryabchikova et al., A&A, 2020, 642, A101



CONCLUSION

- ▶ **Fundamental parameters of stars are determined by high-resolution spectroscopy. The derived abundances are typical for Ap stars and are characterized mainly by excesses up to an order of magnitude of atmospheric iron peak element abundances and very significant excesses up to 3-4 dex of the rare-earth elements in comparison with the solar values.**
- ▶ **We consider possible influence of the surface chemical inhomogeneity of HD 220825 on the derived fundamental parameters through the observed spectral energy distribution (SED) and showed that surface flux variability corresponds to ± 100 K effective temperature variations which is a typical error for Ap stars.**
- ▶ **Comparison of fundamental parameters of HD 220825 and HD 120198 with the interferometric results shows that the model with spectroscopic parameters better fits the hydrogen line profiles.**
- ▶ **And although spectroscopy better identifies fundamental parameters, photometry can be used for statistical research.**

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THANKS FOR YOUR ATTENTION!